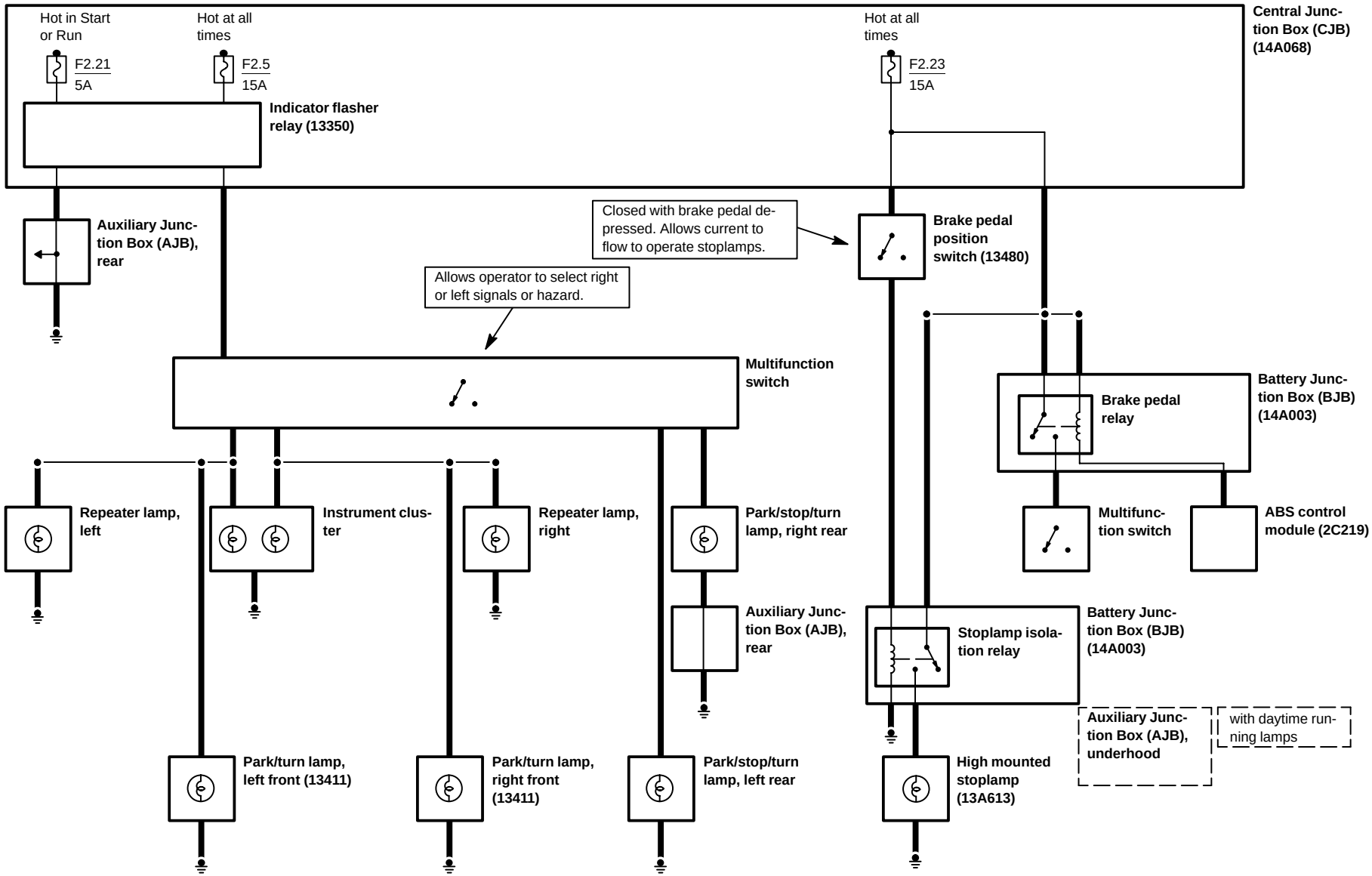with IVD, System overview